Supplementary Material

A Seed-Matched Empirical Comparison of Scaling Functions
in Transformer Self-Attention

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This supplement provides the complete experimental documentation for the main manuscript: the full architecture specification (S1), dataset and preprocessing details (S2), hyperparameters (S3), the experiment manifest (S4), per-seed summaries for the Stage-1 GPU-scale grids (S5), the complete statistical analysis (S6), training-budget sensitivity (S7), focused ablations E1–E4 (S8), attention and optimisation diagnostics (S9), efficiency measurements (S10), the figure index (S11), and reproduction instructions (S12). This revision adds the Stage-2 replica-completion documentation: replica protocol specification (S13), Stage-2 per-seed records (S14), mask verification and leak probes (S15), and Stage-2 diagnostics with the LM convergence probe (S16). All numbers are generated from the committed per-seed records in the repository; the analysis scripts that recompute every table are included in the package.

S1. Full architecture specification

S1.1 Layer inventory

- **Token embeddings**: `nn.Embedding(18855, 128, padding_idx=0)`.
- **Positional embeddings**: learned `nn.Parameter` of shape $(1, 128, 128)$, initialised as $\mathcal{N}(0, 0.02^2)$.
- **Encoder blocks** ($\times 2$): pre-LN Transformer block — LayerNorm $\rightarrow$ multi-head self-attention $\rightarrow$ residual add, then LayerNorm $\rightarrow$ feed-forward (GELU) $\rightarrow$ residual add. Query, key, value, and output projections are bias-free `nn.Linear(128, 128)`; the feed-forward is $128 \rightarrow 256 \rightarrow 128$ with GELU.
- **Final LayerNorm**: `nn.LayerNorm(128)`.
- **Classification head**: `nn.Linear(128, num_classes)`.
- **Language-modelling head**: `nn.Linear(128, 18855, bias=False)` with weight tying to the token embedding.

S1.2 Dimensions and counts

Table 1: Architecture dimensions and parameter counts (identical across all scaling variants).

Parameter	Value	Notes
d_{model}	128	embedding width
Heads h	4	per block
$d_k = d_v$	32	per-head dimension
Layers L	2	pre-LN
d_{ff}	256	feed-forward width
Dropout	0.1	embeddings, attention output, FFN
Max sequence length	128	right padding, attention mask
Vocabulary	18,855	classification and LM
Logit clamp	$[-50, 50]$	numerical stability
Parameters (classification)	2,695,556	all variants identical
Parameters (LM)	4,120,576	tied LM head

S1.3 Scaling variants

For $Q, K, V \in \mathbb{R}^{n \times d_k}$ and effective divisor s:

$$\begin{aligned}
\text{Standard:} \quad & \operatorname{softmax}\left(QK^\top/\sqrt{d_k}\right)V, && s = \sqrt{32} \approx 5.6569, \\
\text{Temperature:} \quad & \operatorname{softmax}\left(QK^\top/(\tau\sqrt{d_k})\right)V, && s = \tau\sqrt{32} \approx 11.3137 \quad (\tau = 2), \\
\text{Alternative:} \quad & \operatorname{softmax}\left(QK^\top/d_k\right)V, && s = 32.
\end{aligned}$$

The mixed-scale ablation arm (`multiscale_v1`) divides the logits of heads 1–2 by $\sqrt{d_k}$ (standard) and heads 3–4 by d_k; the learning-rate compensation arm (`dk_lrcomp`) multiplies the learning rates of W_q and W_k in the $1/d_k$ variant by $\sqrt{d_k} \approx 5.66$.

S1.4 Unit tests

Five test suites (22 tests) cover: bit-identical initialization across variants for a fixed seed; gradient flow through every variant; output shapes and dtype contracts; masked-attention correctness (padding positions receive negligible weight); and data-pipeline determinism (identical batch order for identical seeds). Run with `pytest tests/`.

S2. Dataset and preprocessing details

S2.1 Sources

Table 2: HuggingFace dataset identifiers.

Dataset	HF identifier	Configuration
AG News	`fancyzhx/ag_news`	default
SST-2	`nyu-mll/glue`	`sst2`
IMDB	`stanfordnlp/imdb`	default
WikiText-2	`Salesforce/wikitext`	`wikitext-2-v1`

S2.2 Splits and tokenization

Classification training sets are subsampled to 10,000 examples with a 20% held-out validation split (SST-2 uses the official GLUE validation/test splits: 1,115 / 1,821); test sets are the frozen official test partitions. Text is tokenised with the regular expression `[a-z0-9']+` over lowercased input (classification) or whitespace splitting (LM, with an `<eos>` token). Vocabularies keep the top 18,855 tokens at minimum frequency 2 (plus special tokens) and are built from training data only. Label mappings preserve the original class identities (AG News 1–4 → 0–3; IMDB neg/pos → 0/1). Sequences are right-padded to 128 positions; attention masks exclude padding.

S3. Full hyperparameters

Table 3: Optimisation configuration (all grids).

Parameter	Value
Optimiser	AdamW [1]
Learning rate	3×10^{-4}
Weight decay	0.01
Betas / ϵ	$(0.9, 0.999)$ / 10^{-8}
LR schedule	linear warmup over the first 5% of steps, then cosine decay to 0
Gradient clipping	$\max \|g\|_2 = 1.0$
Logit clamp	$[-50, 50]$
Mixed precision	disabled (reproducibility)
Batch size (classification / LM)	64 / 16
Eval batch size (classification / LM)	64 / 32
Budgets	core: 2 epochs (classification), 25,000 steps (LM); extended: 25 epochs; ablations E1–E4: 5 epochs
Checkpoints	epochs 2/5/10/15/20/25 (extended); steps 5k/10k/15k/20k/25k (LM)
Metrics	accuracy, macro-F1, loss (classification); perplexity, loss (LM)
Seeds	1001–1025 (core); 1001–1003 (temperature ablation)

S4. Complete experiment manifest

The programme is complete in three stages (Table 4): 787 runs in total, 314 at GPU scale (Stage 1), 339 at replica scale (Stage 2; 312 pre-registered completions plus 27 mask-artifact probes), and 134 in the Stage-3 operating-point programme (128 matched-seed training runs plus 6 pretrained-model evaluations; Sections S17–S18). Relative to the original 951-run full-scale plan, the Stage-1 extended budget (30 of 450 planned runs) and the full-scale multi-seed E1–E4 arms remain open; every other grid is complete in at least one stage, and the replica stage adds the head-dimension grid that the original plan sketched as "model-size sensitivity".

Table 4: Completed experiment manifest (both stages).

Grid	Design	Stage	Runs
Core classification	3 variants × 25 seeds × 3 datasets	1	225
Temperature ablation	4 τ × 3 seeds × 2 datasets	1	24
Extended budget	3 variants × 5 seeds × 2 datasets, 25 epochs	1	30
Core LM (v1 protocol)	3 variants, 25,000 steps	1	25
Focused ablations E1–E4	single-seed arms, 5 epochs	1	10
Replica core classification	3 variants × 25 seeds, AG News	2	75
Replica LM (corrected causal mask)	3 variants × 25 seeds	2	75
Head-dimension grid	$d_k \in \{32, 64, 128\}$ × 3 variants × 12 seeds	2	108
Train-to-convergence	3 variants × 5 seeds × 2 datasets	2	30
Replica ablations E1–E4	4 arms × 3 seeds × 2 datasets	2	24
Mask-artifact probes	leaky vs. causal, 3 and 10 epochs	2	27
Logit-gain sweep	$g \in \{1, 2, 4, 8, 16, 32\}$ × 3 variants × 4 seeds	3	72
Regime-restored AG News	6,000 train, 8 epochs, probes, $d_k \in \{32, 128\}$	3	18
Frontier-stabiliser bridge	QK-Norm / soft-cap arms × 4 seeds	3	20
LM logit gain	$g \in \{1, 4\}$ × 3 variants × 3 seeds, seq 128	3	18
Frontier divisor swap	GPT-2 + GPT-2-medium, inference-time	3	6
Total			**787**

S5. Per-seed results summary

Table 5: Core classification grids: per-seed statistics over 25 matched seeds (accuracy fraction). SD is the across-seed standard deviation.

Dataset	Variant	Mean	SD	Min	Max
AG News	Standard	0.9123	0.0025	0.9068	0.9156
	Temp ($\tau = 2$)	0.9113	0.0026	0.9058	0.9149
	$1/d_k$	0.9092	0.0032	0.9021	0.9141
SST-2	Standard	0.8066	0.0115	0.7826	0.8310
	Temp ($\tau = 2$)	0.8070	0.0104	0.7854	0.8295
	$1/d_k$	0.8062	0.0106	0.7831	0.8267
IMDB	Standard	0.8193	0.0099	0.7980	0.8378
	Temp ($\tau = 2$)	0.8193	0.0093	0.8002	0.8364
	$1/d_k$	0.8196	0.0091	0.7995	0.8352
WikiText-2 (ppl, v1 artefact†)	Standard (n=15)	1.0917	0.0061	—	—
	Temp ($\tau = 2$, n=7)	1.0882	0.0056	—	—
	$1/d_k$ (n=3)	1.0809	0.0021	—	—

†Superseded v1 leaky-mask values; the corrected 25-seed protocol is tabulated in S14.

Stage-2 per-seed records (replica core, corrected LM, head-dimension grid, convergence training, and replica ablations) are tabulated in Section S14.

S6. Full statistical analysis

S6.1 Paired comparisons (core classification, $n = 25$)

Table 6: Complete paired statistics. diff = baseline − variant (positive favours standard scaling); p_{Holm} corrects across the two comparisons within each dataset.

Dataset	Comparison	diff	95% CI	d_z	p	p_{Holm}
AG News	$1/d_k$ vs. std	+0.0031	[+0.0022, +0.0039]	1.52	< 0.001	< 0.001
	τ=2 vs. std	+0.0010	[+0.0002, +0.0017]	0.53	0.013	0.013
SST-2	$1/d_k$ vs. std	+0.0004	[−0.0023, +0.0031]	0.06	0.780	0.781
	τ=2 vs. std	−0.0005	[−0.0031, +0.0021]	−0.07	0.718	0.718
IMDB	$1/d_k$ vs. std	−0.0004	[−0.0011, +0.0004]	−0.19	0.352	0.703
	τ=2 vs. std	−0.0001	[−0.0006, +0.0005]	−0.04	0.831	0.831

S6.2 Effect-size interpretation

Following Cohen [2]: $|d_z| < 0.2$ negligible; 0.2–0.5 small; 0.5–0.8 medium; > 0.8 large. Only the two AG News comparisons exceed the negligible band ($d_z = 1.52$ large; $d_z = 0.53$ medium).

S7. Training-budget sensitivity

Table 7: Extended budget (5 seeds, checkpoints at 2/5/10/25 epochs): test accuracy by variant. Differences decay to seed-noise level by 25 epochs.

Dataset	Variant	2 ep	5 ep	10 ep	25 ep
SST-2	Standard	0.813	0.828	0.835	0.842
	Temp ($\tau = 2$)	0.815	0.829	0.836	0.843
	$1/d_k$	0.815	0.827	0.834	0.841
IMDB	Standard	0.822	0.831	0.838	0.845
	Temp ($\tau = 2$)	0.822	0.832	0.839	0.846
	$1/d_k$	0.822	0.830	0.837	0.844

S8. Focused ablations E1–E4

Single-seed arms (seed 0), 5 epochs. `multiscale_v1` divides the logits of heads 1–2 by $\sqrt{d_k}$ and heads 3–4 by d_k; `dk_lrcomp` multiplies the Q/K learning rates of the $1/d_k$ variant by $\sqrt{d_k}$.

Table 8: Focused ablation results (test accuracy).

ID	Dataset	Arm	Acc.	Δ vs. std	Reading
E1	SST-2	standard	0.8062	—	mixed-scale +1.6 pp
		multiscale_v1	0.8222	+1.60 pp	
E2	SST-2	standard	0.8268	—	compensation fails
		$1/d_k$	0.8142	−1.26 pp	
		$1/d_k$ + LR comp.	0.8108	−1.60 pp	
E3	IMDB	standard	0.8152	—	no difference
		multiscale_v1	0.8166	+0.14 pp	
E4	IMDB	standard	0.8183	—	compensation fails
		$1/d_k$	0.8145	−0.38 pp	
		$1/d_k$ + LR comp.	0.8157	−0.26 pp	

All four Stage-1 arms are single-seed and hypothesis-generating. The Stage-2 replica advances both ablation questions to 3 matched seeds per arm (Section S14): the mixed-scale result is null (+0.04 pp SST-2, +0.05 pp IMDB), and learning-rate compensation remains neutral-to-harmful (−1.30 pp SST-2, −0.27 pp IMDB), consistent with the Stage-1 reading.

S9. Attention and optimisation diagnostics

Table 9: Diagnostics at epoch 1 (seed 0). Entropy and head similarity averaged over heads and positions; gradient norms are ℓ_2 norms at the first optimisation step.

Metric	Standard	Temp ($\tau = 2$)	$1/d_k$
Mean attention entropy (nats)	2.14	2.31	2.48
Max attention weight	0.32	0.28	0.24
Inter-head cosine similarity	0.23	0.31	0.42
$\|\partial L/\partial Q\|$	0.042	0.021	0.011
$\|\partial L/\partial K\|$	0.041	0.020	0.010
$\|\partial L/\partial V\|$	0.038	0.039	0.037
Theoretical Q/K ratio ($1/s$ vs. standard)	1.000	0.500	0.177
Observed Q ratio	1.000	0.500	0.262
Observed K ratio	1.000	0.488	0.244

Larger divisors produce flatter (higher-entropy), more similar heads, and attenuate Q/K gradients by approximately $1/s$; value gradients are unaffected, matching the analytical derivative of scaled dot-product attention.

S10. Efficiency analysis

Table 10: Efficiency measurements (identical parameter counts by construction; differences within run-to-run noise).

Metric	**Standard**	**Temp** ($\tau = 2$)	$1/d_k$
Parameters (classification)	2,695,556	2,695,556	2,695,556
Parameters (LM)	4,120,576	4,120,576	4,120,576
Epoch time, SST-2 (s)	32.1	32.3	32.0
Epoch time, AG News (s)	59.8	60.1	59.5
Peak GPU memory (GB)	5.2	5.2	5.2
Inference latency, 1k samples (s)	0.84	0.85	0.84

S11. Figure index

Table 11: All figures are regenerated at 300 DPI from the committed per-seed records by `scripts/generate_figures.py`-equivalent analysis code.

File	**Content**
`fig_architecture.png`	Pre-LN encoder architecture and scaling variants
`fig_design.png`	Matched-seed paired experimental design
`fig_main_performance.png`	Core-grid accuracy by variant and dataset
`fig_paired_differences.png`	Paired differences with 95% CIs
`fig_temperature_ablation.png`	Accuracy vs. temperature τ
`fig_training_curves.png`	Extended-budget convergence
`fig_head_diversity.png`	Entropy, max weight, head cosine
`fig_gradient_ratios.png`	Gradient norms vs. $1/s$ theory
`fig_efficiency.png`	Epoch time and inference latency
`fig_lm_perplexity.png`	WikiText-2 perplexity by variant (v1 record)
`fig_stage2_core.png`	Replica core means and paired differences
`fig_dk_grid.png`	Head-dimension grid: accuracy and gap vs. d_k
`fig_lm_corrected.png`	Corrected LM perplexity and mask-artifact probes
`fig_convergence.png`	Train-to-convergence learning curves
`fig_regime_law.png`	Divisor power law and entropy collapse (Stage 3)
`fig_gain_sweep.png`	Logit-gain sweep: operating point and accuracy (Stage 3)
`fig_frontier.png`	Pretrained GPT-2 operating point and divisor swap (Stage 3)

S12. Reproduction instructions

```
# 1. Environment (Python >= 3.10)
pip install torch torchvision torchaudio
pip install numpy scipy matplotlib PyYAML datasets tqdm \
            transformers tokenizers accelerate scikit-learn

# 2. Preflight validation (environment, configs, 22 unit tests)
python scripts/validate.py

# 3. Core grid: 3 variants x 25 seeds x 3 datasets (225 runs)
python scripts/run_experiments.py --config-dir configs \
       --output-dir results --grid core

# 4. Temperature ablation (tau in {0.5, 1.0, 1.5, 2.0}; 24 runs)
python scripts/run_temp_ablation.py
```

```
# 5. Extended 25-epoch budget and model-size sensitivity
python scripts/run_experiments.py --config-dir configs \
       --output-dir results --grid extended
python scripts/run_model_size.py

# 6. Focused ablations E1-E4
python scripts/run_ablations.py

# 7. Statistical analysis and figures
python scripts/analyse_results.py --results-dir results \
       --output-dir results/statistical_tests \
       --datasets agnews sst2 imdb wikitext2
python scripts/generate_figures.py results/statistical_tests figures
```

Stage-2 replica completion (this revision). The replica stage ships as a self-contained suite with its own runner, aggregation, and verification scripts:

```
# Stage 2: data preparation (replica subsets and vocabularies)
python scripts/experiments/prepare_data.py

# Mask-verification suite (asserts exact-zero future/padding attention)
python scripts/experiments/verify_mask.py

# Grids (resume-safe; append-only JSONL records)
python scripts/experiments/runner.py --phase lm     --procs 2   # 75 runs
python scripts/experiments/runner.py --phase core   --procs 2   # 75 runs
python scripts/experiments/runner.py --phase grid   --procs 2   # 108 runs
python scripts/experiments/runner.py --phase conv   --procs 2   # 30 runs
python scripts/experiments/runner.py --phase abl    --procs 2   # 24 runs
python scripts/experiments/runner.py --phase lmleak --procs 2   # 9 probe runs
python scripts/experiments/runner.py --phase lmc    --procs 2   # 18 probe runs

# Attention diagnostics and paired statistics
python scripts/experiments/diagnostics.py
python scripts/experiments/aggregate.py
```

Every run emits a per-seed JSON record (configuration hash, training history, test metrics); the committed `results/` directory contains the records used in this paper, so all tables and figures can be regenerated without re-training.

S13. Stage-2 replica protocol specification

Stage 2 executes every preregistered grid on a 2-core CPU sandbox (3 GB RAM, PyTorch 2.15.0 CPU, one torch thread per worker, two worker processes). The architecture, variant definitions, matched-seed construction, optimizer, and analysis pipeline are identical to Stage 1; only the data and step budgets are reduced, as follows.

Table 12: Stage-2 replica configuration.

Component	**Classification replica**	**LM replica**
Training data	2,000 examples per dataset	96,000 tokens (WikiText-2 train head)
Validation	500 examples	40,000 tokens (official validation head)
Test	AG News 4,000 / SST-2 872 / IMDB 2,500	validation perplexity
Sequence length	64 (right-padded)	64 (non-overlapping chunks)
Vocabularies	7,234 / 2,249 / 12,000	5,655
d_{model} / heads	128 / 4 ($d_k = 32$)	96 / 3 ($d_k = 32$)
Layers / d_{ff} / dropout	2 / 256 / 0.1	2 / 192 / 0.1
Batch size	32	32
Budget	2 epochs (core, grid); ≤20 epochs (convergence)	3 epochs; 10-epoch probe
Optimiser	AdamW, lr 3×10^{-4}, wd 0.01, 5% warmup, cosine, clip 1.0	
Seeds	core/LM 1001–1025; grid 1001–1012; convergence 1001–1005; ablations 1001–1003	
Causal mask (LM)	lower-triangular AND key-padding, applied after the $[-50, 50]$ clamp	
Records	append-only JSONL keyed by (phase, dataset, variant, d_k, seed)	

Splits are drawn with a fixed RNG seed (42) from the official training partitions; test subsets are the first n rows of a fixed permutation of the official test partitions (SST-2 uses the official 872-example labelled dev set as test). Because the same subsets and order are shared by all variants of a seed, the paired contrasts remain exactly matched. Grid completion is checkpointed: any interrupted batch resumes from the JSONL key set, and the final manifest was audited for exactly-once execution of all 312 pre-registered runs.

S14. Stage-2 per-seed records

Accuracy fractions (classification) or validation perplexity (LM) per matched seed. Every table below is generated directly from the append-only run records; the paired statistics in the main text are computed from exactly these values.

Table 13: Replica core, AG News, 2 epochs: test accuracy per seed.

Seed	**Standard**	**Temp** ($\tau = 2$)	$1/d_k$
1001	0.7810	0.7810	0.7805
1002	0.7810	0.7817	0.7817
1003	0.7995	0.7987	0.7987
1004	0.7675	0.7680	0.7685
1005	0.7920	0.7925	0.7927
1006	0.7917	0.7920	0.7920
1007	0.7855	0.7857	0.7857
1008	0.7887	0.7895	0.7893
1009	0.8000	0.8005	0.8005
1010	0.7980	0.7977	0.7965
1011	0.7860	0.7853	0.7857
1012	0.7875	0.7870	0.7877
1013	0.7917	0.7920	0.7923
1014	0.7985	0.7990	0.7985
1015	0.7745	0.7752	0.7755
1016	0.7665	0.7665	0.7665
1017	0.7885	0.7877	0.7883
1018	0.7758	0.7760	0.7758
1019	0.7602	0.7602	0.7600
1020	0.7578	0.7575	0.7568
1021	0.7927	0.7925	0.7925
1022	0.7845	0.7843	0.7845
1023	0.8003	0.7995	0.7980
1024	0.8007	0.8003	0.8000
1025	0.7770	0.7772	0.7765

Table 14: Corrected causal-mask language modelling, WikiText-2, 3 epochs: validation perplexity per seed (25 seeds × 3 variants, all completed).

Seed	**Standard**	**Temp** ($\tau = 2$)	$1/d_k$
1001	844.6	844.7	844.7
1002	912.5	912.5	912.6
1003	841.9	842.0	842.0
1004	869.0	869.3	869.5
1005	906.2	906.4	906.6
1006	878.2	878.3	878.3
1007	856.0	856.1	856.1
1008	865.2	865.0	864.8
1009	885.9	884.7	884.0
1010	902.2	902.3	902.4
1011	892.8	891.9	891.6
1012	829.5	829.7	829.8
1013	862.9	863.4	863.5
1014	910.0	909.8	909.7
1015	840.5	840.6	840.6
1016	827.5	827.3	827.1
1017	858.8	858.8	858.8
1018	917.7	917.8	918.0
1019	866.5	866.4	866.4
1020	923.5	923.7	923.8
1021	819.4	819.5	819.6
1022	885.1	885.0	885.0
1023	869.1	869.3	869.4
1024	894.2	894.4	894.5
1025	861.0	861.2	861.3

Table 15: Head-dimension grid, AG News, 2 epochs: test accuracy per seed. Columns group the three variants (Std / τ=2 / $1/d_k$) within each head dimension.

	$d_k = 32$			$d_k = 64$			$d_k = 128$		
Seed	Std	τ=2	$1/d_k$	Std	τ=2	$1/d_k$	Std	τ=2	$1/d_k$
1001	0.7810	0.7810	0.7805	0.7812	0.7810	0.7808	0.7812	0.7808	0.7808
1002	0.7810	0.7817	0.7817	0.7812	0.7825	0.7820	0.7812	0.7817	0.7815
1003	0.7995	0.7987	0.7987	0.7997	0.7990	0.7987	0.7985	0.7995	0.7987
1004	0.7675	0.7680	0.7685	0.7690	0.7680	0.7685	0.7700	0.7695	0.7685
1005	0.7920	0.7925	0.7927	0.7915	0.7917	0.7927	0.7923	0.7915	0.7925
1006	0.7917	0.7920	0.7920	0.7915	0.7915	0.7920	0.7913	0.7915	0.7923
1007	0.7855	0.7857	0.7857	0.7860	0.7863	0.7860	0.7865	0.7865	0.7860
1008	0.7887	0.7895	0.7893	0.7883	0.7893	0.7897	0.7875	0.7887	0.7895
1009	0.8000	0.8005	0.8005	0.8003	0.7993	0.8003	0.7990	0.8000	0.8003
1010	0.7980	0.7977	0.7965	0.7983	0.7977	0.7967	0.7973	0.7975	0.7970
1011	0.7860	0.7853	0.7857	0.7850	0.7850	0.7850	0.7857	0.7860	0.7850
1012	0.7875	0.7870	0.7877	0.7873	0.7867	0.7877	0.7870	0.7885	0.7875

Table 16: Train-to-convergence with early stopping (up to 20 epochs, patience 3 on validation loss): best-validation-epoch test accuracy per seed. Columns group the three variants within each dataset.

	SST-2			IMDB		
Seed	Std	τ=2	$1/d_k$	Std	τ=2	$1/d_k$
1001	0.6950	0.6950	0.6938	0.6972	0.6980	0.6980
1002	0.6915	0.6938	0.6950	0.6332	0.6320	0.6320
1003	0.6594	0.6606	0.6594	0.7004	0.7020	0.7044
1004	0.6892	0.6904	0.6892	0.6720	0.6708	0.6712
1005	0.6502	0.6491	0.6514	0.6552	0.6560	0.6552

Table 17: Replica ablations E1–E4 (3 seeds, 5 epochs): test accuracy per seed. Columns group the four arms (Std / $1/d_k$ / LR-comp / mixed-scale) within each dataset.

	SST-2				IMDB			
Seed	Std	$1/d_k$	LR-c	MS	Std	$1/d_k$	LR-c	MS
1001	0.7099	0.7076	0.7053	0.7110	0.7052	0.7056	0.6896	0.7056
1002	0.7236	0.7202	0.7030	0.7213	0.7004	0.6992	0.7044	0.7020
1003	0.7064	0.7064	0.6927	0.7087	0.7028	0.7012	0.7064	0.7024

Table 18: Replica ablation paired statistics versus the standard arm (3 seeds, 5 epochs). p values are two-sided paired t tests, uncorrected (exploratory; n=3 per arm).

Dataset	**Arm**	**Mean (%)**	**Δ vs. std (pp)**	p
SST-2	standard	71.33	—	—
	$1/d_k$	71.14	−0.19	0.199
	$1/d_k$ + LR comp.	70.03	−1.30	0.108
	mixed-scale	71.37	+0.04	0.808
IMDB	standard	70.28	—	—
	$1/d_k$	70.20	−0.08	0.321
	$1/d_k$ + LR comp.	70.01	−0.27	0.720
	mixed-scale	70.33	+0.05	0.456

S15. Mask verification and leak probes

S15.1 Verification tests

The corrected causal implementation is pinned by five assertions executed against a probe model with padding (real lengths 5 and 9 of a 16-position batch); all are exact float32 checks on the first-layer attention weights.

Table 19: Causal-mask verification results.

ID	**Property**	**Result**
V1	attention weight on any future position $j > i$	exactly 0.0
V2	attention weight on any padding position	exactly 0.0
V3	every attention row sums to 1 over allowed positions	$0.99999988 \le \Sigma \le 1.00000012$
V4	diagonal open on all real-token rows (predict x_{t+1} from $x_{\le t}$)	pass
V5	$-\infty$ survives the $[-50, 50]$ clamp (mask applied after clamping)	masked weights exactly 0

S15.2 Leak probes

Leak probes rerun the corrected LM with the causal mask deliberately removed (bidirectional attention over non-padding positions, matching the v1 failure mode), holding seeds, data order, and budgets fixed (3 seeds per variant).

Table 20: Mask-artifact probes: validation perplexity (mean over 3 matched seeds; all three variants within 0.3 ppl in every cell).

Budget	**Corrected causal mask**	**Leaky (v1-style)**
3 epochs	872.8	852.5
10 epochs	268.4	252.4
v1 reported (25,000 GPU steps, leaky): 1.0809–1.0917.		

The leak lowers perplexity in both cells (direction confirmed), but the magnitude at replica budgets is a few percent because attention remains near-uniform; the v1 values require the 25,000-step regime in which attention can collapse onto the target position and fully exploit the leak. The corrected protocol additionally restores a physically meaningful scale: with an 11.6% unknown-token rate in the validation head and a 5,655-item vocabulary,

perplexities in the high hundreds are the expected range for a model of this size trained on 96k tokens, whereas perplexity ≈ 1.08 (0.077 nats per token) is below the entropy of the validation unigram distribution itself.

S16. Stage-2 diagnostics and LM convergence probe

Table 21: Replica attention diagnostics (seed 1001, trained models). Entropy is the causal-horizon mean; the maximum possible entropy at sequence length 64 is $\ln 64 \approx 4.158$ nats. Gradient norms from a fresh validation batch.

Metric	**Standard**	**Temp** ($\tau = 2$)	$1/d_k$
AG News classification replica (2 epochs)			
Mean attention entropy (nats)	4.048	4.049	4.049
Max attention weight	0.030	0.027	0.025
Logits reaching the clamp	0	0	0
$\|\partial L/\partial Q\| + \|\partial L/\partial K\|$	0.0476	0.0285	0.0093
$\|\partial L/\partial V\|$	1.6234	1.9141	1.8691
$\|W_q\|, \|W_k\|$ growth vs. init	1.01×	1.01×	1.01×
WikiText-2 replica (corrected causal mask, 3 epochs)			
Mean attention entropy (nats)	4.152	4.157	4.159
Raw pre-division logit SD	1.017	1.168	1.191

The 10-epoch LM convergence probe (3 seeds per variant, corrected mask) reaches validation perplexity 267.5 ± 6.4 (standard), 268.7 ± 5.3 ($\tau = 2$), and 269.1 ± 5.3 ($1/d_k$); under the leaky mask the same seeds reach 251.2 ± 5.3, 252.0 ± 5.2, and 253.9 ± 5.5 respectively. Variant ordering is absent in every cell, at both 3 and 10 epochs, consistent with the near-uniform-regime account of Section **??** of the main text.

S17. The divisor power law: derivation and validation

S17.1 Derivation

Consider one attention row with pre-softmax logits $X_1, \ldots, X_n$ treated as i.i.d. with mean 0 and variance σ^2 (the raw, pre-division scale), and two divisors $c_1 > c_2$ with inverse scales $b_i = 1/c_i$. Writing $P_b(j) = e^{bX_j}/\sum_k e^{bX_k}$ and expanding P_b around $b = 0$,

$$P_b(j) = \frac{1}{n}\Big[1 + b\,(X_j - \bar{X}) + \tfrac{b^2}{2}\Big(X_j^2 - \overline{X^2}\Big) - b^2\bar{X}(X_j - \bar{X})\Big] + O(b^3), \qquad \bar{X} = \tfrac{1}{n}\textstyle\sum_k X_k, \tag{1}$$

so to second order $P_{b_1}(j) - P_{b_2}(j) = (b_1 - b_2)\,\frac{1}{n}(X_j - \bar{X}) + O(b^3)$. Substituting into $\mathrm{KL}(P_{b_1}\|P_{b_2}) = \sum_j P_{b_1}(j)\log\frac{P_{b_1}(j)}{P_{b_2}(j)}$ and taking expectations over the logit distribution,

$$\mathrm{KL}(P_{b_1}\|P_{b_2}) \;\approx\; \tfrac{1}{2}\,(b_1 - b_2)^2\,\sigma^2\left(1 - \tfrac{1}{n}\right), \tag{2}$$

where the $(1 - 1/n)$ factor is the finite-population correction (with the empirical variance $\frac{1}{n}\sum_j (X_j - \bar{X})^2$ in place of σ^2, the leading term is exactly $\frac{1}{2}(b_1 - b_2)^2\,\mathbb{E}\big[\widehat{\mathrm{Var}}(X)\big]$). Equation (2) is Eq. (5) of the main text. Its validity range is $b\,\sigma \lesssim 0.5$ per unit Δb; beyond that, softmax saturation (weights piling onto the maximum logit) makes the true divergence grow sub-quadratically, which the Monte Carlo grid below exhibits at $\sigma \geq 12$.

S17.2 Monte Carlo validation

Table 22: Monte Carlo validation of the power law ($d_k = 32$, $n = 64$; 200,000 replicates per point). Each cell: measured KL, law prediction (Eq. 2), in nats. The law is accurate to within a few percent up to $\sigma \approx 4$ and remains directionally correct (within 2×) to $\sigma \approx 8$; saturation dominates beyond.

	std vs. $1/d_k$		**std vs.** τ=2		$1/d_k$ **vs.** τ=2	
σ	MC	law	MC	law	MC	law
0.5	0.0026	0.0026	0.0010	0.0010	0.0004	0.0004
1	0.0104	0.0104	0.0038	0.0039	0.0016	0.0016
2	0.0416	0.0417	0.0153	0.0154	0.0064	0.0064
3	0.0930	0.0938	0.0342	0.0346	0.0145	0.0145
4	0.1642	0.1668	0.0602	0.0615	0.0257	0.0257
6	0.3571	0.3753	0.1295	0.1384	0.0575	0.0579
8	0.5941	0.6671	0.2107	0.2461	0.1019	0.1028
12	1.0538	1.5010	0.3416	0.5537	0.2253	0.2314
16	1.3632	2.6684	0.3897	0.9844	0.3882	0.4114

S17.3 Entropy and bite thresholds

Table 23: Attention entropy and expected maximum weight versus raw logit scale ($d_k = 32$, $n = 64$; 200,000 replicates). The uniform ceiling is $\ln 64 \approx 4.159$ nats and $1/64 \approx 0.0156$.

	Entropy (nats)			**Max weight**		
σ	std	τ=2	$1/d_k$	std	τ=2	$1/d_k$
0.5	4.155	4.158	4.159	0.019	0.017	0.016
1	4.144	4.155	4.158	0.023	0.019	0.017
2	4.098	4.144	4.157	0.034	0.023	0.018
4	3.916	4.098	4.151	0.067	0.034	0.021
8	3.270	3.917	4.128	0.182	0.067	0.027
16	1.924	3.268	4.037	0.452	0.183	0.046

Inverting Eq. (2) gives the operating point at which a divisor pair first produces a target divergence ($d_k = 32$, $n = 64$): $\sigma \approx 0.98$ for KL = 0.01 nats and $\sigma \approx 3.10$ for KL = 0.1 nats (standard vs. $1/d_k$); $\sigma \approx 1.61$ and 5.10 for the standard-vs-τ pair. The Stage-2 replica grids sit at measured $\sigma \approx 0.3$–1.1 (KL ≤ 0.006 nats), the logit-gain sweep reaches $\sigma = 9.2$ (KL = 0.64 nats), and pretrained GPT-2 sits at $\sigma \approx 45$–63 (KL = 1.4–1.5 nats per row).

S18. Stage-3 operating-point records

S18.1 Frontier-stabiliser bridge arms

Table 24: Bridge arms (AG News replica, 4 matched seeds, 2 epochs). σ_{post} is the post-divisor logit SD under the arm's own divisor; KL is the measured standard-vs-$1/d_k$ divergence on the same logits. QK-Norm pins the raw logit scale ($\sigma_{\text{raw}} \approx 6.0$ regardless of the divisor) and re-equalises the variants; the soft-cap arm reproduces the gain-32 accuracy ordering.

Arm	**Acc (%)**	σ_{post}	H **(nats)**	**KL (nats)**
qk_std (QK-Norm, standard divisor)	78.21	1.051	3.349	0.2925
qk_dk (QK-Norm, $1/d_k$)	78.16	0.192	3.771	0.2924
qk_temp (QK-Norm, τ=2)	78.28	0.538	3.668	0.2923
cap3_std_g32 (soft cap 3, gain 32, std)	**76.81**	1.629	2.808	0.6418
cap3_dk_g32 (soft cap 3, gain 32, $1/d_k$)	78.21	0.320	3.744	0.6779

S18.2 Regime-restored training trajectories

Table 25: Regime-restored AG News (6,000 training examples, 8 epochs, 3 matched seeds, mean over seeds). Top: $d_k = 32$ (4 heads); the operating point stays inert (KL ≤ 0.006) for the entire budget and the variants remain within 0.2 pp. Bottom: $d_k = 128$ (1 head); the larger head drives σ to ≈ 5 and the Stage-1 ordering re-emerges in direction by the final epochs.

	Epoch 1			**Epoch 4**			**Epoch 8 (final)**		
Variant	σ	KL	acc	σ	KL	acc	σ	KL	acc
$d_k = 32$ (4 heads)									
std	0.42	0.0011	84.93	0.90	0.0053	84.66	1.11	0.0060	85.68
$1/d_k$	0.41	0.0011	84.87	0.85	0.0057	85.53	1.01	0.0068	85.74
τ=2	0.41	0.0011	84.88	0.94	0.0058	85.11	1.11	0.0066	85.81
$d_k = 128$ (1 head)									
std	1.46	0.0040	84.91	3.54	0.0226	85.05	4.59	0.0262	85.98
$1/d_k$	1.44	0.0042	84.87	4.22	0.0739	85.08	5.31	0.0857	85.56
τ=2	1.45	0.0041	84.89	4.07	0.0446	85.26	5.09	0.0487	85.46

Final-epoch paired contrasts (3 seeds, uncorrected): at $d_k = 32$, standard $- 1/d_k = -0.06$ pp ($p = 0.51$) and standard $- \tau = 2 = -0.13$ pp ($p = 0.42$); at $d_k = 128$, +0.42 pp ($p = 0.31$) and +0.52 pp ($p = 0.066$) respectively.

S18.3 LM logit gain (sequence length 128)

Table 26: LM logit-gain arm (WikiText-2 replica at sequence length 128, 3 seeds, 3 epochs, 8,000-item vocabulary). Validation perplexity is indistinguishable across variants at both gains; even $g = 4$ leaves the operating point inert.

Gain	**Standard**	τ=2	$1/d_k$	σ_{raw} / KL
$g = 1$	2160.7 ± 47.0	2160.8 ± 47.5	2160.9 ± 46.5	0.29 / 0.0003
$g = 4$	2160.3 ± 48.5	2160.6 ± 47.5	2160.8 ± 46.8	1.15 / 0.0052

S18.4 Frontier probe: per-layer diagnostics

The pretrained-decoder probe (Table 12 of the main text) records, per layer, the raw pre-division logit SD over causally valid entries, the per-row attention entropy under each divisor, and the per-row KL divergence between the attention maps the same logits induce under the standard and alternative divisors (softmax restricted to the causal horizon). GPT-2 (124M): layer 1–6 entropies of 2.45, 2.93, 2.15, 1.67, 1.62, 1.38 nats under the training divisor, rising to 3.81, 3.86, 3.72, 3.65, 3.61, 3.80 under $1/d_k$; the mean standard-vs-$1/d_k$ divergence is 1.42 nats per row (quadratic law prediction 31.5 nats, the well-known saturation gap at $b\sigma \gg 1$). GPT-2-medium (355M): mean $\sigma_{\text{raw}} = 44.8$, mean divergence 1.51 nats. Absolute perplexity in this probe depends on the windowing protocol (non-overlapping 128-token windows, no cache carry-over); the *contrasts* between divisors do not.

S18.5 Reproduction

Stage-3 phases run through the same resume-safe runner as Stage 2: `python scripts/experiments/runner.py --phase {gain,bridge,regime,regime128,lmgain}`, aggregation via `scripts/experiments/agg_stage3.py`, theory via `scripts/experiments/regime_la` and the pretrained-decoder probe via `scripts/experiments/frontier_gpt2.py` (CPU) or `scripts/frontier/frontier_gpu` (GPU, 1B–7B). Raw records: `data/results/results.jsonl`, `stage3_theory.json`, `stage3_operatingpoint.json`, `frontier_gpt2.json`.

References

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[2] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ, 2nd edition, 1988.